MINH BUI TRAN CONG

Senior Applied AI Engineer

Agentic AI | Technical Leadership | Enterprise Solution Delivery

Ho Chi Minh City, Vietnam
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AI engineer with 5 years of experience and technical leadership across enterprise AI systems. Evaluate and apply agentic workflows, knowledge graphs and document AI to automate business processes and improve software lifecycle traceability. Translate stakeholder requirements into architecture, guide experimentation and code reviews, and coordinate delivery with attention to context quality, resource use and operational reliability.

PROFESSIONAL EXPERIENCE

FPT Software AI Center — Sep 2024 - Present

Senior AI Engineer

Provided technical leadership for an 8-9-person team across system design, experimentation and code reviews; clarified stakeholder requirements and coordinated frontend/backend delivery.

Enterprise contract intelligence & agent architecture

- Designed the AI architecture for an enterprise contract-analysis platform serving approximately 1,000 users, using LangGraph to coordinate multi-step document review and tool-based analysis.
- Made retrieved answers traceable to source documents through citation-backed retrieval; added response caching to reuse LLM outputs and evaluated lower-cost models against task requirements.
- Preserved contract metadata and master-data history across revisions through snapshot versioning, supported by PostgreSQL/pgvector and OpenSearch data models.

Enterprise integration & feature access

- Integrated the contract platform with an internal AI service, addressing cross-service authentication and secure connectivity requirements within enterprise infrastructure.
- Designed Salesforce Canvas and SSO integration to embed the contract platform into existing business workflows, defining application embedding and API contracts across systems.
- Enabled business-unit-specific feature access through a Business Line permission matrix, enforcing the same feature restrictions across backend APIs and the application UI.

Knowledge graphs & software lifecycle intelligence

- Built traceable links between requirements, documentation, source code and defects in a knowledge graph, giving QA and engineering a connected view of information spread across Rally, wikis and code repositories.
- Led a feasibility study for test generation grounded in requirement dependencies; modeled epic, feature and user-story relationships to supply context for scenarios spanning multiple requirements.

Document intelligence & applied research

- Designed ingestion to preserve document layout and section hierarchy for contextual retrieval; evaluated vision-language classification and document parsers to inform the pipeline design.

PROFESSIONAL EXPERIENCE / CONTINUED

Spartan

Oct 2023 - Dec 2024

AI Engineer

Document extraction, workflow APIs & cloud delivery

- Researched document-parsing approaches and led PDF parser development with Python libraries and Azure Document Intelligence, preserving layout for downstream extraction.
- Applied cloud-based LLMs to automate structured data extraction and formatting, connecting PDF parsing to reusable document-processing workflows.
- Designed LLM invocation scheduling and rate-limit handling to manage constrained API capacity across concurrent document-processing workloads.
- Led backend development for a LangFlow-based application that lets users compose PDF processing workflows through a visual interface; built APIs for flow execution and modular parsing components.
- Contributed Kubernetes deployment scripts and supported GCP rollout, working across backend integration and deployment troubleshooting.

Truong Minh Thinh Technology JSC

Jul 2021 - Oct 2023

AI Engineer

Vietnamese NLP, model optimization & conversational applications

- Designed and maintained scripted chatbot interaction flows; integrated LangChain and prompt-based responses to handle open-ended queries beyond predefined scenarios.
- Developed Vietnamese question-generation models using BARTPho and Marian, including custom pointer mechanisms for out-of-vocabulary handling.
- Built sentiment-classification models for user messages to support analysis and visualization of chatbot interactions.
- Fine-tuned pretrained language models using Hugging Face SFT Trainer and LoRA, with task-specific hyperparameter experimentation.
- Applied ONNX and TorchScript for inference optimization; used DVC to version data and experiments and keep model iterations reproducible.

TECHNICAL EXPERTISE

Agents & retrieval: LangGraph, LangChain, tool calling, RAG, knowledge graphs, semantic chunking and retrieval workflows.

Backend & data: Python, FastAPI, PostgreSQL, pgvector, OpenSearch, Redis, SQLAlchemy, Alembic, REST APIs, SSE, snapshot versioning.

Enterprise integration: Salesforce Canvas, SSO integration design, Microsoft Graph, cross-service authentication, Business Line feature flags and API/UI access controls.

Models & document AI: PyTorch, Hugging Face Transformers, SFT, LoRA, ONNX, TorchScript, DVC, Azure Document Intelligence, AWS Textract.

Application delivery: Docker, Kubernetes, GCP, Flutter, automated backend tests, requirements discovery, architecture, technical reviews and cross-team delivery.

SELECTED ENGINEERING PROJECTS

Multi-Agent Executive Assistant

LangGraph | FastAPI | PostgreSQL | Microsoft Graph | Docker

- Developed a stateful multi-agent assistant routing requests across task management, news briefings, meeting summaries, email drafting and document search.
- Implemented LLM relevance filtering for news and extracted report structure from sample PDF/DOCX files so generated briefings follow the requested format; supported PDF and DOCX export.
- Worked on PostgreSQL checkpoint connection lifecycle, async task persistence and deadline/status handling; implemented human-in-the-loop confirmation flows for Teams/Planner integration.

PRISM | AI Self-Reflection App

Internal QA

Flutter | FastAPI | PostgreSQL | Redis | RevenueCat

- Developed a bilingual iOS product and async Python backend with provider-agnostic AI responses streamed over server-sent events (SSE).
- Implemented Redis-backed cost reservations and cancellation cleanup to control concurrent generation spend and release resources after interrupted requests.
- Worked on subscription webhook reconciliation and account-merge consistency, alongside encrypted private text and privacy-conscious telemetry; prepared release checks and local validation.

LangChain Chatbot

OpenAI / Vertex AI | MongoDB | Gradio | LangSmith | Microsoft Presidio

- Built a streaming chatbot with MongoDB-backed conversation memory and configurable bot behavior; added LangSmith prompt versioning and tracing.
- Integrated Microsoft Presidio to anonymize personally identifiable information before sending content to the LLM.

github.com/minhbtrc/langchain-chatbot

EDUCATION

Master of Computer Science

Aug 2025 - Present

Ho Chi Minh University of Technology

Bachelor of Computer Science

Aug 2018 - Apr 2023

Ho Chi Minh University of Technology

PROFESSIONAL DEVELOPMENT

AI Agents Fundamentals - Hugging Face (Feb 2025)
Develop Your AI Skills with Google Gemini and Google Cloud Platform - LinkedIn (Dec 2024)
Generative AI with Large Language Models - Coursera (Aug 2023)
Deep Neural Networks with PyTorch - IBM (Mar 2022)
Natural Language Processing Specialization - Coursera (Mar 2022)